

Project FORESIGHT: AI-Powered Demand Forecasting & Inventory Risk Management

1. Executive Summary

Project FORESIGHT is an AI-driven inventory intelligence system developed to forecast product demand, identify inventory risks, and recommend business actions for improving operational efficiency.

The system uses machine learning algorithms to predict future sales demand while analysing inventory levels to identify Stockout Risk and Overstock Risk. It helps businesses make better purchasing and replenishment decisions, reducing inventory costs and improving customer satisfaction.

2. Business Objective

The objectives of this project are:

- Forecast future product demand accurately.
- Minimize stockout situations.
- Reduce excess inventory.
- Improve inventory planning.
- Increase business profitability.
- Support data-driven business decisions.

3. Revenue at Risk

Revenue at Risk represents the estimated sales that may be lost because products are unavailable when customers want to purchase them.

Business Impact

- Lost sales opportunities
- Reduced customer satisfaction
- Lower customer retention
- Negative impact on business growth

Formula

$$\text{Revenue at Risk} = \text{Predicted Demand} \times \text{Selling Price}$$

Recommendation

- Increase replenishment frequency.
- Improve demand forecasting.
- Monitor fast-moving products daily.

4. Capital Locked

Capital Locked represents the amount of money invested in inventory that is not selling quickly.

Business Impact

- Increased warehouse costs
- Higher holding costs
- Reduced cash flow
- Lower inventory turnover

Formula

$$\text{Capital Locked} = \text{Overstock Quantity} \times \text{Unit Cost}$$

Recommendation

- Offer discounts on slow-moving products.
- Bundle products.
- Reduce future purchase quantities.
- Monitor inventory ageing regularly.

5. Inventory Risk Analysis

The project classifies inventory into the following categories:

a. High Stockout Risk

Products expected to run out of stock based on forecasted demand.

Recommended Action:

- Reorder immediately
- Increase safety stock
- Monitor daily

b. Medium Risk

Inventory levels are acceptable but should be monitored.

Recommended Action:

- Weekly inventory review
- Maintain reorder schedule

c. Overstock Risk

Products with excess inventory.

Recommended Action:

- Reduce future procurement
- Launch promotional offers
- Apply markdown pricing

6. Machine Learning Model Performance

The project evaluates multiple forecasting models using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Square Error)
- MAPE (Mean Absolute Percentage Error)
- WAPE (Weighted Absolute Percentage Error)
- R^2 Score

The forecasting model is also compared against a Seasonal Naive Baseline to ensure improved forecasting accuracy.

7. Recommended Business Actions

a. For High Demand Products

- Increase purchase quantity.
- Maintain higher safety stock.
- Review demand daily.

b. For Slow-Moving Products

- Reduce inventory levels.
- Launch promotional campaigns.
- Bundle products with fast-selling items.

c. For Overstock Products

- Apply markdown pricing.
- Reduce procurement.
- Improve inventory rotation.

d. For Stockout Products

- Trigger automatic reorder alerts.
- Increase reorder quantity.
- Improve supplier coordination.

8. Expected Business Benefits

- Improved forecast accuracy
- Reduced inventory costs
- Lower stockout frequency
- Better cash flow management
- Improved inventory turnover
- Higher customer satisfaction
- Increased operational efficiency
- Better strategic decision-making

9. Conclusion

Project FORESIGHT provides an intelligent demand forecasting and inventory management solution that helps organisations optimise stock levels, minimise business risks, and improve profitability. By combining machine learning, inventory analytics, and business intelligence, the system enables proactive decision-making and supports efficient supply chain management.